This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.

PHYS 312	**Modern Physics**	**3(3,0)**
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and particle physics.		

II. College Requirements

A. Remedial Courses

Proficiency in the English Language is a requirement for admission to any program in the College of Arts and Sciences. Students who do not meet the University English proficiency admission requirements may gain admission through the Intensive English Program, in which they can develop their English proficiency. In addition, incoming students in the Computer Science, Healthcare Information Systems, and Mathematics programs must pass the mathematics placement exam. Those who fail the mathematics placement exam and obtain a low score must take one or two remedial Mathematics courses before taking program-level calculus courses. The possible remedial courses are:

- MATH 190 - Calculus I
- MATH 191 – Calculus II

These remedial courses do not count toward fulfilling the degree requirements. A description of the remedial courses follows.

MATH 190	**Calculus I**	**3(3,0)**
Functions and their graphs, limits and continuity, differentiation, applications of derivatives, Extreme values and mean value theorem, definite and indefinite integrals, and the substitution method.		

MATH 191	**Calculus II**	**3(3,0)**
Covers limits and indeterminate forms with L'Hôpital's Rule, optimization, antiderivatives, and the Fundamental Theorem of Calculus. Topics include substitution methods, areas, volumes (cross-sections, shells), arc length, and surfaces of revolution. It introduces parametric and vector functions with applications to motion, arc length in space, curvature, and normal vectors. **Prerequisite**: Math 190.		

B. Mandatory Courses

All students are required to take nine credit hours of mandatory foundational courses. A list of these courses and their descriptions follows.

Course	Title	Credits	Prerequisite
ARAB 212	Arabic Language and Communication	2	
	Community & Sustainability	1	
ENGL 210	English Composition and Rhetoric	3	
ENGL 217	Professional English Communication	3	

Descriptions of these courses are given below.

ARAB 212	**Arabic Language and Communication**	**2(2,0)**
This course helps students develop their ability to communicate effectively in standard Arabic. It provides students with the necessary Arabic communication skills they may need in their future jobs. Specifically, students learn how to write and orally present different forms of workplace documents in Arabic.		

CMNS 200	**Etiquette**	**1(1,0)**
This course is designed to help students develop interpersonal and communication skills that are fundamental to success in the workplace, no matter what industry, organization, or sector they are employed in. Students will improve their professional style as they study topics including polite conversation, personal appearance, office politics, diplomacy, telephone, cell phone, and voicemail etiquette, meeting protocol, job interview presentations, and even international travel. Students will participate in an off-campus formal dining experience. **Co-requisite:** ENGL 210.		

CMNS 205	**First Aid**	**1(1,0)**
This course provides an overview of the principles of first aid, including preserving life, preventing further injury, and protecting the unconscious. It enhances students' knowledge of first-aid procedures and their application in real-life situations. The course also helps them develop the skills necessary to assist until medical help arrives. It raises students' awareness of different types of medical emergencies and the appropriate steps to take in both mild and life-threatening conditions. **Co-requisite:** ENGL 210.		

ENGL 210	**English Composition and Rhetoric**	**3(3,0)**
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, research papers, and oral presentations. **Prerequisite**: ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	**Professional English Communication**	**3(3,0)**
This course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails,		

memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.

III. Program Requirements

Requirements for the bachelor's degree are program-specific. They encompass two categories: Major and non-major. Each category consists of mandatory and elective courses. The program requirements for the bachelor's degree in the various CAS majors are provided below. Details and titles of relevant courses are included in the Student's Study Plan (SSP), which every CAS student will have.

Course Coding

Each course offered by the College of Arts and Sciences is designated by a four-letter code representing the program or elective category, followed by a three-digit number denoting the course number; for example, "**EDUC 301**."

The four-letter codes below refer to the respective course categories as follows:

ADPR: Advertising and Public Relations
ARAB: Arabic
CMNS: Communications
COSC: Computer Science
EDUC: Education
ENGL: English Language
GRDS: Graphic Design
HCIS: Healthcare Information Systems
HIST: History
IELP: Intensive English Language Program
INDS: Interior Design
JRSM: Journalism – Media and Digital Communication
MATH: Mathematics
POSC: Political Science
PSYC: Psychology
SOCI: Sociology

The first digit in the three-digit course number indicates the course level: **1** for freshman, **2** for sophomore, **3** for junior, **4** for senior, and **5** for graduate.

The following example demonstrates the coding system: EDUC 201 is a sophomore-level education course.

The designation used to represent the credit-hour breakdown (t, p) of a course is as follows: "c" is the total credit hours, "t" is the theoretical component of the course, and "p" is the practical or laboratory component. For example, 3(3, 0) denotes a 3-credit-hour course with 3 contact lecture hours and 0 laboratory hours.

Learning Support Center

The Learning Support Center (LSC) at the College of Arts and Sciences is another manifestation of RHU's resolute commitment to students' academic success. The LSC offers the students a peer-to-peer learning opportunity that complements classroom learning. The Center aims to support students in advancing their knowledge and skills, reinforcing what they learn in the classroom or overcoming learning and performance difficulties. In addition to serving as a focal point for inquiry and two-way "asker-replier" learning, the LSC provides space for academic interactions that can significantly enhance learning. The Center is in Block I and is open to all students throughout the week.

The LSC is staffed by Teaching Assistants, upper-level undergraduate students, who are always available to assist their fellow students in overcoming academic challenges. Additionally, "on-call" faculty members are assigned to ensure that the LSC is acting in the best interests of the students. The center's resources include books, study guides, course materials, boards, computers, large-screen LCD panels, and internet-connected PCs.